

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                                                            | Mark                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 1a              | <p>A description that includes six of the following points:</p> <ul style="list-style-type: none"> <li>• dependent variable is mass needed to break fibres (1)</li> <li>• use fibres the same length and {diameter/cross sectional area} (1)</li> <li>• method of supporting fibre (1)</li> <li>• add known mass to fibre until it breaks (1)</li> <li>• suitable method of control of one variable {age of fibre / temperature/ humidity} (1)</li> <li>• method of calculation for tensile strength (1)</li> <li>• repeats and calculate a mean (1)</li> </ul> | <p>Accept thickness</p> <p>Accept between 2 (clamp stands) / with fibre hung from one (stand)</p> <p>Accept use of forcemeter – record force when it breaks</p> <p>Accept age of plant<br/>Accept AC room for temperature (not waterbath)<br/>AC room / humidifier / dehumidifier for humidity</p> <p>eg force divided by (cross sectional) area / mass converted to force then divided by (cs) area</p> <p>Accept repeats to calculate SD</p> | <p><b>Exp</b><br/><b>6</b></p> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                       | Additional Guidance                                                                                                     | Mark                           |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>1b</b>       | <p>A description that includes three of the following points:</p> <ul style="list-style-type: none"> <li>cellulose {is a polymer of / contains many /contains a large number of} B glucose (molecules) (1)</li> <li>joined by 1-4 glycosidic bonds (1)</li> <li>every other (glucose) molecule is inverted (1)</li> <li>to give a straight (chain / molecule) (1)</li> </ul> | <p>Accept polysaccharide containing beta glucose molecules</p> <p>Accept reversed</p> <p>Accept linear / unbranched</p> | <p><b>Exp</b><br/><b>3</b></p> |

| Question Number | Answer                                                                                                                                                                                        | Additional Guidance                                                                                                                                                                             | Mark             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1c              | <p>An answer that includes the following:</p> <ul style="list-style-type: none"> <li>• biofuels only release the carbon dioxide that was absorbed by {photosynthesis / plants} (1)</li> </ul> | <p>Accept they are (almost) carbon neutral</p> <p>Accept biofuels release carbon dioxide which plants would have released when they decompose / biofuels don't produce extra carbon dioxide</p> | <p>Exp<br/>1</p> |

(Total for question 1 = 10 marks)